

ASSIGNMENT 10

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
KD3_98823_pranoti@localhost> DELIMITER $$
KD3_98823_pranoti@localhost> CREATE PROCEDURE sp_rectangle(l_length INT,w_width INT)
-> BEGIN
-> declare v_area INT;
-> declare v_perimeter INT;
-> SET v_area =l_length *w_width;
-> SET v_perimeter =2*(l_length +w_width);
->
-> select v_area as area;
-> select v_perimeter as perimeter;
->
-> end;
-> $$
Query OK, 0 rows affected (0.01 sec)

KD3_98823_pranoti@localhost> DELIMITER ;
KD3_98823_pranoti@localhost> call sp_rectangle(4,4);
+-----+
| area |
+-----+
|    16 |
+-----+
1 row in set (0.00 sec)

+-----+
| perimeter |
+-----+
|         16 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
KD3_98823_pranoti@localhost> DELIMITER $$
KD3_98823_pranoti@localhost> CREATE PROCEDURE sp_num()
-> BEGIN
-> DECLARE num INT;
-> DECLARE v_value INT;
-> DECLARE v_square INT;
-> DECLARE v_cube INT;
-> SET num = 5;
-> SET v_value = num;
-> SET v_square = num * num;
-> SET v_cube = num * num * num;
->
-> INSERT INTO Temp VALUES (v_value, v_square, v_cube);
-> SELECT v_value AS 'val', v_square AS 'sqr', v_cube AS 'cub';
-> END;
-> $$
```

Query OK, 0 rows affected (0.03 sec)

```
KD3_98823_pranoti@localhost>
KD3_98823_pranoti@localhost> DELIMITER ;
KD3_98823_pranoti@localhost> CALL sp_num();
```

```
+-----+-----+-----+
| val  | sqr  | cub  |
+-----+-----+-----+
|    5 |   25 |  125 |
+-----+-----+-----+
1 row in set (0.03 sec)
```

Query OK, 0 rows affected (0.03 sec)

```
KD3_98823_pranoti@localhost> SELECT * FROM Temp;
+-----+-----+-----+
| val  | sqr  | cub  |
+-----+-----+-----+
|    5 |   25 |  125 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
KD3_98823_pranoti@localhost> DELIMITER $$
KD3_98823_pranoti@localhost> create procedure sp_temp(IN C FLOAT ,IN F FLOAT)
-> BEGIN
-> DECLARE v_F FLOAT;
-> DECLARE v_C FLOAT;
-> set v_F=(9/5)*C +32;
-> SET v_C =(F-32)*(5/9);
-> SELECT v_F AS Fahrenheit;
-> SELECT v_C AS Celsius;
-> END;
-> $$
Query OK, 0 rows affected (0.03 sec)

KD3_98823_pranoti@localhost> DELIMITER ;
KD3_98823_pranoti@localhost> call sp_temp(20,30);
+-----+
| Fahrenheit |
+-----+
|          68 |
+-----+
1 row in set (0.02 sec)

+-----+
| Celsius |
+-----+
| -1.1112 |
+-----+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.02 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
-> ^C
KD3_98823_pranoti@localhost> DELIMITER $$
KD3_98823_pranoti@localhost>
KD3_98823_pranoti@localhost> CREATE PROCEDURE sp_convert(IN total_inches INT)
-> BEGIN
->     DECLARE v_yards INT;
->     DECLARE v_feet INT;
->     DECLARE v_inches INT;
->
->     SET v_yards = total_inches / 36;
->     SET v_inches = total_inches % 36;
->     SET v_feet = v_inches / 12;
->     SET v_inches = v_inches % 12;
->
->     SELECT v_yards AS Yards,
->            v_feet AS Feet,
->            v_inches AS Inches;
-> END $$
Query OK, 0 rows affected (0.03 sec)

KD3_98823_pranoti@localhost> DELIMITER ;
KD3_98823_pranoti@localhost> CALL sp_convert(120);
+-----+-----+-----+
| Yards | Feet | Inches |
+-----+-----+-----+
|      3 |     1 |       0 |
+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
KD3_98823_pranoti@localhost> DELIMITER $$
KD3_98823_pranoti@localhost> CREATE PROCEDURE sp_product(IN num1 FLOAT, IN num2 FLOAT)
-> BEGIN
->     DECLARE v_product FLOAT;
->     SET v_product = num1*num2;
->     IF v_product >= 100 THEN
->     SELECT 'product is equal to or greater than 100' AS result;
->     ELSE
->     SELECT 'product is less than 100' AS result;
->     end if;
-> end;
-> $$
Query OK, 0 rows affected (0.03 sec)

KD3_98823_pranoti@localhost>
KD3_98823_pranoti@localhost> DELIMITER ;
KD3_98823_pranoti@localhost> CALL v_product(10,11.5);
ERROR 1305 (42000): PROCEDURE practice.v_product does not exist
KD3_98823_pranoti@localhost> CALL sp_product(10,11.5);
+-----+
| result |
+-----+
| product is equal to or greater than 100 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

